Experiment No. 8

Aim: Design and Verification of Half Adder using logic gates.

Requirement: Digital IC Trainer Kit, Power Supply and Connecting Wires.

Procedure/ Design Steps:

1. Implement the truth table for half adder
2. Simply the output using K-map
3. Implement the circuit using trainer kit
4. Verify the truth table of all the outputs (sum and carry)

Theory:

An adder is a digital logic circuit in electronics that implements addition of numbers. In many computers and other types of processors, adders are usedto calculate addresses, similar operations and table indices in the ALU and also in other parts of the processors. These can be built for many numerical repre- sentations like excess-3 or binary coded decimal. Adders are classified into two types: half adder and full adder. The half adder circuit has two inputs: A and B, which add two input digits and generate a carry and sum. The full adder circuit has three inputs: A and C, which add the three input numbers and generate a carry and sum. This article gives brief information about half adder and full adder in tabular forms and circuit diagrams.

Half Adder

An adder is a digital circuit that performs addition of numbers. The half adderadds two binary digits called as augend and addend and produces two outputsas sum and carry; XOR is applied to both inputs to produce sum and AND gate is applied to both inputs to produce carry. Truth table for half and full adder is shown in Table 12

The circuit implementation for half adder and full adder is shown in Figures 12.

$$S = \overline{A} \cdot B + A \cdot \overline{B} = A \oplus B$$

$$C = A \cdot B$$

Inputs			Outputs	
A	B	CIN	COUT	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Table 22: Truth Table for Half Adder

Diagram:

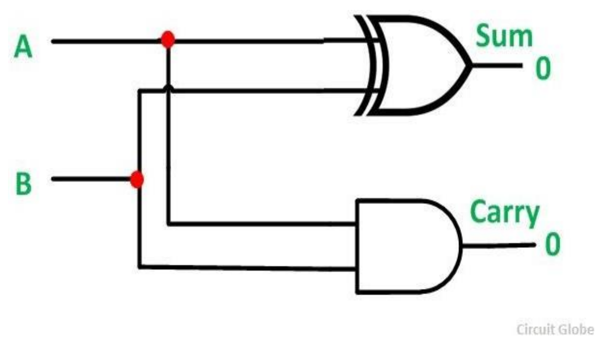

Figure 19: Half adder

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Viva - Voce Questions 6.1

Question 6.1 *Design 2 bit adder using full adder*